Refined sugar (white, dextrose): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified standard for Refined sugar (white, dextrose) (Sweeteners and Syrups), stated as sold. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; face value, no reconstitution; read-across / control / not-material for unregulated cells; no occurrence percentiles)

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	100	government maximum level
Cadmium (Cd)	150	government maximum level
Total arsenic (tAs)	100	government maximum level
Inorganic arsenic (iAs)	100	bounded by tAs (reflex speciation)

Mercury, total (tHg)	50	government maximum level
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	800	human-grade read-across (European Union)
Tin (Sn)	250,000	human-grade read-across (Singapore)
Aluminium (Al)	100,000	human-grade read-across (China)
Total chromium (Cr)	1,000	human-grade read-across (China)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	100	government maximum level
Cadmium (Cd)	150	government maximum level
Total arsenic (tAs)	100	government maximum level
Inorganic arsenic (iAs)	100	Inorganic arsenic is a component of total arsenic (iAs <= tAs). The iAs value (500 ug/kg) is not below the binding total arsenic ceiling (100 ug/kg), so inorganic arsenic is governed by and bounded by the total arsenic ceiling; no separate iAs ceiling is published. It is enforced by reflex speciation whenever tAs is elevated.
Mercury, total (tHg)	50	government maximum level
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	800	European Union — human-food analog
Tin (Sn)	250,000	Singapore — human-food analog
Aluminium (Al)	100,000	China — human-food analog

Total chromium (Cr)	1,000	China — human-food analog
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

Pendergrass, K. (2026). *Refined sugar (white, dextrose): HMTc standard* (Standard No. 10.83138/hmtc.std.31.refined-sugar). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.31.refined-sugar

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References

[1] China — human-food maximum levels
Binds Al, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/refined-sugar. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

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